Supplementary Material

Multimodal Integration of Clinical, Transcriptomic, and Epigenomic Features for Breast Cancer Molecular Subtyping

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August 2026

S1. Complete Feature Engineering Details

S1.1 Clinical Feature Encoding

Table 1 lists the 22 clinical variables from the METABRIC clinical_patient_wide and clinical_sample_wide CSVs, their encoding method, and resulting feature dimensions.

Table 1: Clinical variables and their encoding.

Variable	Type	Encoding	Dimensions
AGE_AT_DIAGNOSIS	Continuous	Z-score	1
TUMOR_SIZE	Continuous	Z-score	1
LYMPH_NODES_EXAMINED_POSITIVE	Continuous	Z-score	1
NPI	Continuous	Z-score	1
MUTATION_COUNT	Continuous	Z-score	1
TMB_NONSYNONYMOUS	Continuous	Z-score	1
OS_MONTHS	Continuous	Z-score	1
RFS_MONTHS	Continuous	Z-score	1
BREAST_SURGERY	Categorical (2)	One-hot	2
CELLULARITY	Categorical (3)	One-hot	3
CHEMOTHERAPY	Categorical (2)	One-hot	2
CLAUDIN_SUBTYPE	Categorical (7)	One-hot	7
ER_IHC	Categorical (2)	One-hot	2
HER2_SNP6	Categorical (4)	One-hot	4
HISTOLOGICAL_SUBTYPE	Categorical (8)	One-hot	8
HORMONE_THERAPY	Categorical (2)	One-hot	2
INFERRED_MENOPAUSAL_STATE	Categorical (2)	One-hot	2
LATERALITY	Categorical (2)	One-hot	2
RADIO_THERAPY	Categorical (2)	One-hot	2
ER_STATUS	Categorical (2)	One-hot	2
HER2_STATUS	Categorical (2)	One-hot	2
PR_STATUS	Categorical (2)	One-hot	2
GRADE	Categorical (4)	One-hot	4
ONCOTREE_CODE	Categorical (8)	One-hot	8
Total			**69**

S1.2 Feature Dimensions Across Modalities

Table 2: Feature dimensions for each modality.

Modality	Raw Dimensions	Final Dimensions
Clinical	22 variables	69
mRNA expression	412 genes	412
Copy number alteration	373 genes	256
DNA methylation	232 probes	232
Somatic mutations	173 genes	173
Total (all 5)		**1,142**
Selected 3		**713**

S2. Comprehensive Model Comparison

Table 3 provides the complete model comparison results including all evaluated algorithms, hyperparameter settings, and feature combinations.

Table 3: Complete model comparison with 5-fold cross-validation.

Model	**Features**	**CV Acc (%)**	**Std**
Majority class	–	35.6	–
Single-modality models			
LogReg C=0.01	Clinical (69d)	64.1	2.7
LogReg C=0.01	mRNA (412d)	73.8	1.6
LogReg C=0.01	Methylation (232d)	54.7	2.4
Multi-modality models			
LogReg C=0.01	Clin+mRNA (481d)	74.5	1.9
LogReg C=0.01	Clin+Methyl (301d)	65.7	1.9
LogReg C=0.01	mRNA+Methyl (644d)	73.1	2.6
LogReg C=0.01	**Clin+mRNA+Methyl (713d)**	**75.3**	**2.6**
LogReg C=0.01	Clin+mRNA+CNA (737d)	73.0	1.8
LogReg C=0.01	Clin+mRNA+Mutat (654d)	73.3	2.0
LogReg C=0.01	All 5 (1142d)	72.4	2.0
Regularization sweep (Clin+mRNA+Methyl, 713d)			
LogReg C=0.001	713d	73.9	2.7
LogReg C=0.01	713d	75.3	2.6
LogReg C=0.1	713d	74.1	2.3
LogReg C=1.0	713d	72.9	2.1
LogReg C=10.0	713d	72.4	1.9
Calibrated LogReg	713d	74.4	2.1
Tree-based models (Clin+mRNA+Methyl, 713d)			
Random Forest	713d	72.4	2.4
XGBoost	713d	72.0	3.3
LightGBM	713d	71.0	3.1
CatBoost	713d	73.9	2.7
Extra Trees	713d	71.3	1.7
Ensemble methods (713d)			
Soft-vote LR+GB+XGB	713d	72.6	2.8
Soft-vote LR+RF+GB	713d	72.5	2.8
Transformer models (713d)			
LightTransformer (5-mod)	713d	68.5	1.8
TF (3-mod, d=64, 2L)	713d	68.5	1.8
TF (3-mod, d=128, 3L)	713d	70.7	2.1
Hybrid TF prob + LR	713d	74.8	2.7
TF-AE + LogReg	713d	73.7	2.0
Guided Transformer	713d	74.0	1.0
LR → Guided TF Cascade	713d	76.2	2.0

S3. Feature Selection Analysis

ANOVA F-test was used to rank features by discriminative power. Figure **??** shows the cross-validation accuracy as a function of the number of top-ranked features for mRNA-only and

mRNA+clinical+methylation models.

Key observations:

- mRNA-only performance plateaus at approximately 200 genes (74.5% CV)
- With clinical+methylation, 200 mRNA genes achieve 75.5% CV (slightly better than full 412 genes at 74.7%)
- Top discriminating mRNA genes: *ACOT4* (F=130.8), *A2ML1* (F=102.0), *ABAT* (F=98.5), *ACTR3B* (F=95.0)

Table 4: Feature selection results for different top-k values.

Model	**Top-k genes**	**Total dims**	**CV Acc (%)**
mRNA only	50	50	72.9 ± 2.1
mRNA only	100	100	73.5 ± 2.3
mRNA only	200	200	74.5 ± 2.1
mRNA only	412 (all)	412	73.8 ± 1.6
Clin+mRNA+Methyl	50	351	73.0 ± 2.4
Clin+mRNA+Methyl	100	401	74.0 ± 2.2
Clin+mRNA+Methyl	200	501	75.5 ± 1.2
Clin+mRNA+Methyl	412 (all)	713	75.3 ± 2.6

S4. Luminal A vs. Luminal B Confusion Analysis

The dominant classification error is confusion between Luminal A and Luminal B subtypes (Table 5). These subtypes share ER-positive status and differ primarily in proliferation-related gene expression.

Table 5: Luminal A vs. Luminal B misclassification analysis on the METABRIC test set.

Direction	**Count / Total**	**Rate (%)**
Luminal A → Luminal B	9 / 74	12.2
Luminal B → Luminal A	12 / 50	24.0
Average confidence when misclassified		
P(LumA) when LumB→LumA	0.689	
P(LumB) when LumA→LumB	0.586	

The high confidence score (0.689) when predicting Luminal B as Luminal A suggests the model has difficulty detecting the subtle proliferation signature that distinguishes these subtypes.

S5. McNemar's Statistical Tests

McNemar's test was used to compare the classification decisions of different models on the held-out test set (n=206).

Table 6: McNemar's test comparing LogReg models with different regularization.

Comparison	χ^2	**p-value**
LogReg C=0.01 vs C=0.1	0.083	0.773
LogReg C=0.01 vs C=1.0	0.063	0.803

None of the comparisons reach statistical significance ($p < 0.05$), consistent with the small test set size (n=206). The 5-fold CV results (Table 3) provide more reliable model comparison.

S6. TCGA External Validation Details

S6.1 Data Acquisition

TCGA-BRCA data were obtained via the cBioPortal API. The Firehose Legacy study (brca_tcga) contains 1,108 patients. RNA-seq V2 expression data were available for 1,100 samples, with PAM50 annotations for 987 patients. Clinical and survival data were retrieved separately from the patient-level clinical endpoint.

S6.2 Gene Mapping

All 412 METABRIC mRNA genes were successfully mapped to Entrez Gene IDs. Gene expression values from TCGA RNA-seq V2 were log2-transformed and rank-normalized within the TCGA platform (Equation 1 in the main text).

S6.3 Class Distribution Shift

Table 7: Subtype distribution comparison between METABRIC and TCGA-BRCA.

Subtype	**METABRIC (%)**	**TCGA (%)**
Luminal A	35.7	50.9
Luminal B	23.9	20.2
HER2-enriched	12.5	7.9
Claudin-low	9.5	0.0 (not annotated)
Basal-like	10.4	17.4
Normal-like	7.9	3.6

The substantial distribution shift, particularly the absence of claudin-low annotations and the higher proportion of Luminal A in TCGA, explains part of the macro F1-score drop from 0.703 (METABRIC) to 0.585 (TCGA).

S6.4 TCGA Survival Analysis

On 966 TCGA patients with available survival data (133 death events):

- Overall C-Index: 0.586 (log-rank $p = 0.013$)
- 3-year C-Index: 0.617 (56 events)
- 5-year C-Index: 0.596 (85 events)
- 10-year C-Index: 0.587 (122 events)

S7. SHAP Feature Importance

Table 8 lists the complete SHAP top-20 feature importance rankings.

Table 8: Top 20 features by mean absolute SHAP value.

Rank	Feature	Modality	Mean |SHAP|
1	THREEGENE_ER+/HER2- High Prolif	Clinical	0.0614
2	THREEGENE_ER+/HER2- Low Prolif	Clinical	0.0579
3	ER_STATUS_Positive	Clinical	0.0511
4	ER_STATUS_Negative	Clinical	0.0511
5	ACP7	mRNA	0.0505
6	THREEGENE_ER-/HER2-	Clinical	0.0473
7	ACTG2	mRNA	0.0467
8	ACOT4	mRNA	0.0436
9	ADM	mRNA	0.0425
10	ADCY3	mRNA	0.0416
11	ABCB1	mRNA	0.0407
12	ADRB2	mRNA	0.0374
13	ADORA2B	mRNA	0.0373
14	ACKR1	mRNA	0.0356
15	ACADSB	mRNA	0.0355
16	ACTR3B	mRNA	0.0345
17	ADORA3	mRNA	0.0342
18	ABTB3	mRNA	0.0341
19	ADCY9	mRNA	0.0330
20	ADGRE5	mRNA	0.0321

S8. Comprehensive Results Table

The complete configuration comparison table (24 experiment configurations) is provided as a separate CSV file (`tune_results.csv`).